

SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares total site energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. The construction cost data is from 2026 RSMeans Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	Wall upgrade: • Exterior wall insulation: Fiberglass Batts, target R-20.4 Roof upgrade: • Roof insulation: Blown Mineral Wool, target R-34.5 Window upgrade: • 1-pane replacement • Weatherstrip application (skipped: no OperableWindow)

		- Glazing film application: safety film - Caulking application: acrylic - Window frame replacement: wood window frame Door upgrade: - Door replacement: wooden door (1 of 3 door(s) skipped: door type incompatibility) - Bottom seal: automatic door bottom - Top/side seal: jamb weatherstrip
Scenario 2	wall + roof + window + door	Wall upgrade: - Exterior wall insulation: Extruded Polystyrene (XPS) Foam Board, target R-20.4 Roof upgrade: - Roof insulation: Expanded Polystyrene (EPS) Foam Board, target R-34.5 Window upgrade: 2-pane replacement - Weatherstrip application (skipped: no OperableWindow) - Glazing film application: anti-graffiti film - Caulking application: polyurethane - Window frame replacement: wood-aluminium window frame Door upgrade: - Door replacement: polystyrene core steel door (1 of 3 door(s) skipped: door type incompatibility) - Bottom seal: brush weatherstrip - Top/side seal: silicone adhesive smoke gasket
Scenario 3	wall + roof + window + door	Wall upgrade: - Exterior wall insulation: Polyiso Insulation Foam Board, target R-20.4 Roof upgrade: - Roof insulation: Polyiso Insulation Foam Board, target R-34.5

		Window upgrade: • 3-pane replacement • Weatherstrip application (skipped: no OperableWindow) • Glazing film application: low-e film • Caulking application: polyurethane • Window frame replacement: wood-aluminium window frame Door upgrade: • Door replacement: garage door (skipped due to door type incompatibility) • Bottom seal: automatic door bottom • Top/side seal: silicone adhesive smoke gasket
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Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	440.31	385.63	54.68	12.42%
Scenario 2	Total Site Energy (GJ)	440.31	385.64	54.67	12.42%
Scenario 3	Total Site Energy (GJ)	440.31	385.70	54.61	12.40%

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison

Baseline		\$18,286
Scenario 1		\$17,020

■ Baseline ■ Scenario 1

Max saving: **\$-1,265 (-6.92%)**

Total Site Energy Saving (Scenario - Baseline) (GJ/yr) Comparison

Baseline		440
Scenario 1		386

■ Baseline ■ Scenario 1

Max saving: **55 GJ (12%)**

Min Retrofit Construction Cost

Lowest Retrofit Cost Payback Period

\$20,270

Scenario 1

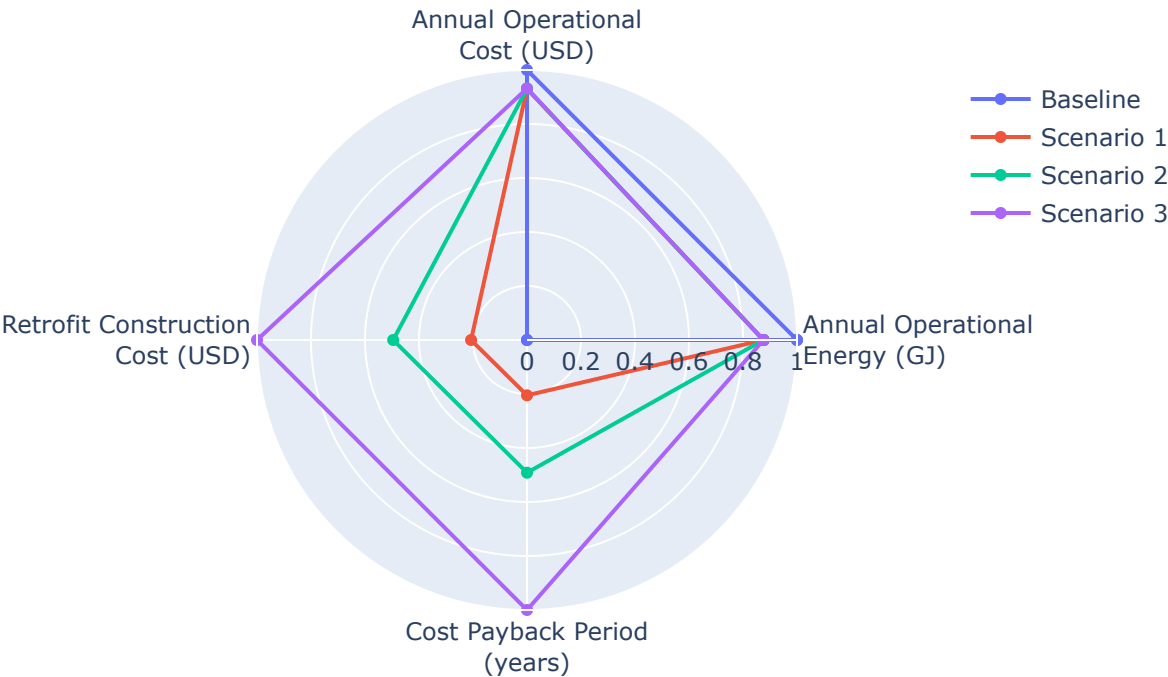
16 years

Scenario 1

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).

Scenario 1		16 yrs
Scenario 2		38 yrs
Scenario 3		78 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m ³)	Area (m ²)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m ³)	Product Lifetime (years)
Scenario 1	Wall insulation	Fiberglass Batts	17	282	N/A	N/A	0.03	30	30
Scenario 1	Roof insulation	Blown Mineral Wool	144	599	0.24	N/A	0.03	90	30
Scenario 1	Window frame	wood window frame	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 1	Window caulking	acrylic	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 1	Door	wooden door	N/A	3.90	N/A	N/A	0.15	638	30
Scenario 1	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 1	Door top/side sealing	jamb weatherstripping	N/A	N/A	N/A	16	N/A	N/A	15
Scenario 2	Wall insulation	Extruded Polystyrene (XPS) Foam Board	17	282	N/A	N/A	0.03	35	30
Scenario 2	Roof insulation	Expanded Polystyrene (EPS) Foam Board	144	599	0.24	N/A	0.03	20	30
Scenario 2	Window frame	wood-aluminum window frame	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 2	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10

Scenario 2	Door	polystyrene core steel door	N/A	3.90	N/A	N/A	0.10	472	30
Scenario 2	Door bottom sealing	brush weatherstripping	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 2	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15
Scenario 3	Wall insulation	Polyiso Insulation Foam Board	13	282	N/A	N/A	0.03	35	30
Scenario 3	Roof insulation	Polyiso Insulation Foam Board	144	599	0.24	N/A	0.03	35	30
Scenario 3	Window frame	wood-aluminum window frame	N/A	6.51	N/A	N/A	N/A	N/A	15
Scenario 3	Window caulking	polyurethane	0.0034	N/A	N/A	N/A	N/A	N/A	10
Scenario 3	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 3	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario breakdown: left chart shows retrofit construction cost contribution by measure, right chart shows total site energy by fuel type (electricity, natural gas, and other). Water is excluded from energy breakdown.

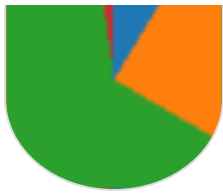
Scenario 1

Cost breakdown by retrofit measure



Total Site Energy Breakdown (GJ)





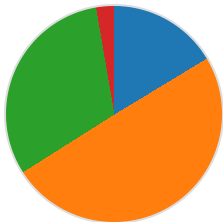
■ Wall: \$1,742 (8.6%)
■ Roof: \$5,021 (24.8%)
■ Window: \$13,082 (64.5%)
■ Door: \$426 (2.1%)
Total: \$20,270



■ Electricity: 369.26 GJ (95.8%)
■ Natural Gas: 16.37 GJ (4.2%)
Total: 385.63 GJ

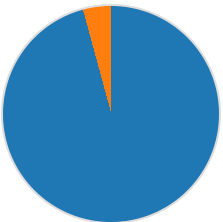
Scenario 2

Cost breakdown by retrofit measure



■ Wall: \$7,919 (16.3%)
■ Roof: \$24,099 (49.7%)
■ Window: \$15,154 (31.3%)
■ Door: \$1,318 (2.7%)
Total: \$48,489

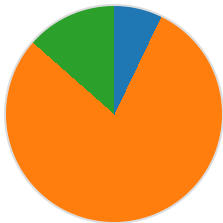
Total Site Energy Breakdown (GJ)



■ Electricity: 369.34 GJ (95.8%)
■ Natural Gas: 16.30 GJ (4.2%)
Total: 385.64 GJ

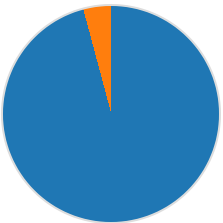
Scenario 3

Cost breakdown by retrofit measure



■ Wall: \$6,959 (7.1%)
■ Roof: \$77,651 (79.4%)
■ Window: \$13,088 (13.4%)
■ Door: \$132 (0.1%)
Total: \$97,831

Total Site Energy Breakdown (GJ)



■ Electricity: 369.61 GJ (95.8%)
■ Natural Gas: 16.08 GJ (4.2%)
Total: 385.69 GJ

Scenario	Retrofit Construction Cost (USD)	Annual Operational Cost (USD)	Total Site Energy (GJ)
Baseline	\$0.00	\$18,286	440.31
Scenario 1	\$20,270	\$17,020	385.63
Scenario 2	\$48,489	\$17,024	385.64
Scenario 3	\$97,831	\$17,034	385.70

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